

Meaning Lies in Structure: Fine-Grained Table-Centric Document Semantic Parsing

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Abstract. Tables encode quantitative evidence that directly informs financial reporting, risk assessment, and policy evaluation. In such contexts, the validity of automated analysis hinges on accurate semantic integration between tabular data and its explanatory text; misalignment can materially distort interpretation and downstream decisions. However, existing PDF table research focuses primarily on visual structure recovery, while fine-grained semantic alignment across document components remains underexplored. Prior work typically adopts coarse binary relevance modeling and fails to capture the hierarchical and heterogeneous nature of table-text relationships. To address this gap, we formulate fine-grained table-text semantic alignment as a five-level hierarchical framework and propose AITABLER, an end-to-end system integrating PDF structural reconstruction, layout-aware spatial context modeling, and LLM-based semantic reasoning. We construct two benchmarks: a fine-grained relevance corpus (29,138 annotated pairs) and a Table-QA dataset (6,000 questions) derived from over 3,000 PDF pages. Experiments demonstrate consistent improvements in relevance grading and substantial gains in downstream Table-QA. Overall, Our method contributes the first fine-grained semantic grading dataset for table-centric document analysis and improves Table-QA performance significantly.

Keywords: Table Understanding · Table Question Answering · Large Language Model.

1 Introduction

Tables constitute one of the most critical information carriers in modern documents, serving as the primary medium for presenting structured data and consolidating key results. In corporate disclosures, government statistics, and industry analyses, decision-critical information is predominantly conveyed through tabular formats [28]. For instance, the U.S. Securities and Exchange Commission (SEC) [20] releases millions of corporate filings annually via its EDGAR

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system [31], where financial data is largely embedded in tables. As the scale of structured reporting and data disclosure continues to expand, automated understanding and analysis of tabular content has evolved from a supplementary tool to a foundational capability, underpinning data-driven decision making and intelligent information processing systems [8,13,26]. Consequently, enabling accurate and scalable interpretation of tables in documents has emerged as a central challenge for both industry and the document intelligence community.

Nevertheless, interpreting tables in real-world documents extends far beyond reading cell values. Essential information for understanding a table, such as metric definitions, experimental settings, assumptions, and explanatory commentary, is frequently located in surrounding paragraphs rather than within the table itself [3,30]. Consequently, accurate interpretation hinges on correctly identifying and integrating the textual context that semantically complements the table. This challenge is further intensified in PDF documents. As a layout-oriented format, PDF preserves visual appearance rather than logical structure [11]; tables, captions, and paragraphs are spatially arranged but not explicitly semantically linked [32]. Reconstructing these relationships therefore becomes a necessary and non-trivial prerequisite for reliable downstream analysis.

Existing research on PDF table analysis has primarily focused on visual processing, including layout detection [32] and structural parsing [15,21]. While effective for reconstructing structure, these approaches remain confined to surface representation and largely overlook semantic integration. A limited number of studies [12] explore relationships between tables and surrounding text; however, such efforts typically adopt coarse-grained relevance modeling that are restricted to binary association or heuristic proximity, lacking explicit characterization of the heterogeneous and hierarchical nature of table-text interactions, which has been shown to be crucial in hybrid table-text question answering and numerical reasoning over documents [3,30,32]. In real-world document workflows, this limitation is substantive rather than superficial. Accurate interpretation frequently depends on precise alignment between tabular data and contextual explanations, where subtle differences in semantic association can materially alter the meaning of quantitative evidence. Without fine-grained modeling of these relationships, automated systems may retrieve incomplete or weakly related context, leading to distorted interpretation and reduced reliability in downstream analytics and decision support.

To address this gap, we propose AITABLER, to the best of our knowledge the first system for fine-grained table-text relevance analysis. AITABLER integrates PDF structural reconstruction with spatial context retrieval and LLM-based fine-grained reasoning to model hierarchical relationships between tables and surrounding text. Building upon this capability, we further develop a semantic-enhanced Table-QA module that enables structure-aware question answering over document tables. Experimental results show that AITABLER achieves over 98% accuracy on strongly related samples, exceeds 75% on weakly related cases, and attains more than 70% accuracy on complex QA tasks, substantially outperforming existing methods. Our contributions are summarized as follows:

- We formulate fine-grained table-text relevance modeling as a structured five-level hierarchical framework, enabling explicit characterization of heterogeneous semantic associations beyond coarse binary relevance.
- We present AITABLER, an end-to-end system that operationalizes this framework through integrated PDF structural reconstruction and multi-level semantic reasoning.
- We curate two large-scale benchmarks: a fine-grained table-text relevance corpus (29,138 annotated pairs) and a purpose-built Table-QA dataset (6,000 questions), derived from over 3,000 pages of real-world PDF documents.
- We provide comprehensive empirical evaluation demonstrating substantial improvements in relevance modeling and downstream QA, and release the AITABLER prototype and datasets to foster reproducible research.⁴

2 Background

2.1 Multi-Level Relevance Grading Scheme

Fine-grained table-text relevance modeling aims to explicitly characterize the diverse semantic associations between table and surrounding narrative content. Despite its importance, there is currently no standardized framework or widely accepted taxonomy for systematically modeling such multi-level relationships.

In this work, we formalize fine-grained alignment as a five-level hierarchical framework that serves as a principled and practically grounded reference design. The hierarchy is guided by two considerations. First, it systematically reflects the three structural components of a table: caption, body-cell content, and row/column headers, through which explicit semantic associations are typically established. Second, it incorporates a residual semantic category to capture relevance beyond these structural facets, thereby ensuring comprehensive coverage of the relevance spectrum. The resulting framework defines a monotonically decreasing relevance ranking over table-text associations.

- **Level 1 (Caption Reference).** The text explicitly or descriptively refers to the table via its caption (e.g., “As shown in Table 3”). Caption reference represents the strongest relevance, as core interpretive discussion in academic writing is typically introduced through such explicit citation.
- **Level 2 (Content Alignment).** The text engages with specific body-cell values, including summarizing trends, comparing entries, or discussing numerical results traceable to the table. Body-cell correspondence is ranked above header overlap because it provides stronger exclusivity and discriminative power: identical data values across different tables are rare, whereas shared headers (e.g., metrics or dataset names) are common.
- **Level 3 (Topical Match).** The text shares conceptual or schema-level overlap with the table’s row/column headers (e.g., dataset names, methods, metrics) without necessarily referencing specific cell values.

⁴ Our code and datasets are available at <https://github.com/xuan084/Meaning-Lies-in-Structure-Fine-Grained-Table-Centric-Document-Semantic-Parsing.git>.

- **Level 4 (Peripheral Context).** A residual semantic category capturing text that is relevant yet does not satisfy the structural criteria of the first three levels, such as background discussion or methodological context.

- **Level 5 (Unrelated).** No discernible semantic relationship with the table.

In sum, Levels 1–3 correspond to the three structural components of a table in decreasing discriminative strength, Level 4 extends coverage through flexible semantic judgment, and Level 5 captures the complement. The design is logically complete that each table-text pair maps to exactly one level and provides a strictly ordered relevance hierarchy that enables fine-grained context modeling. While alternative instantiations are possible, this five-level scheme offers a balanced trade-off between interpretability, coverage, and practical effectiveness.

2.2 Related Works

Table Detection and Structure Recognition. Extensive work has addressed table detection and structure recognition, progressing from CNN-based detectors [19] to transformer architectures [15,21]. Layout-aware pre-training methods such as LayoutLM [25,9] further advance document understanding, supported by large-scale datasets such as PubLayNet [32]. These methods effectively address *where* tables are and *what* they structurally contain, but do not model the semantic relationships between tables and their surrounding text.

Table-Text Semantic Association. Modelling the semantic relationship between tables and their surrounding text remains under-explored. DOTABLER [12] constructed the first table-text association dataset and trained a binary related/unrelated classifier, but the binary formulation does not capture varying degrees of relevance. Multi-task models such as TabPedia [29], along with the broad LLM-based table reasoning capabilities highlighted by recent surveys [26], consume context as a flat input without ranking its relevance. Our work fills this gap with a multi-level grading scheme that directly characterises the semantic relationship between a table and each surrounding text block.

Table Question Answering. Table-QA models have evolved from specialised pre-trained encoders [8,13] to LLM-based methods [22] that achieve strong results without fine-tuning. Benchmarks range from isolated-table settings [18,2] to hybrid table-text scenarios [3,33,30,27,23] that require joint reasoning over tables and surrounding text. However, no existing benchmark controls for *context relevance level* or measures its impact on QA accuracy and token efficiency. Our Table-QA benchmark addresses this gap by pairing every question with relevance-annotated context, enabling the first controlled study of how graded context quality affects downstream performance.

3 Methodology

The overall architecture of AITABLER is illustrated in Figure 1. AITABLER first transforms PDF documents into structured representations via layout detection and OCR (Section. 3.1), and then integrates spatial context modeling with LLM-based fine-grained relational reasoning to capture precise table-text semantic

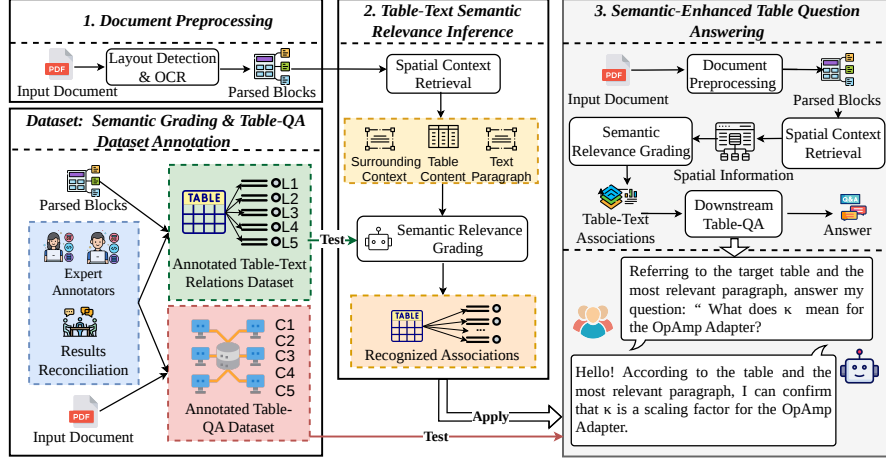


Fig. 1: Overall workflow of AITABLER

associations (Section 3.2). Building upon this capability, we develop a semantic-enhanced Table-QA module that substantially strengthens table-centric question answering through fine-grained relational modeling (Section 3.3). To bridge the research gap, we introduce the first dedicated fine-grained table-text relevance dataset together with a corresponding Table-QA benchmark (Section 3.4).

3.1 Document Preprocessing

Since PDF documents are inherently unstructured and cannot be directly parsed or analyzed, AITABLER first performs a preprocessing step to transform them into structured representations through layout analysis and optical character recognition (OCR). Specifically, we train a document layout detection model based on the PubLayNet [32] dataset, using Faster R-CNN [19] as the backbone architecture. In PubLayNet, document pages are annotated with bounding boxes and categorized into five region types: *Table*, *List*, *Title*, *Text*, and *Figure*. Considering that *List* and *Title* are text-based components and play the same semantic role as regular text in subsequent relationship analysis, we unify *List*, *Title*, and *Text* under a single *Text* category. Based on the trained layout detection model, AITABLER is able to identify each region type on a page and precisely localize its relative spatial position, as represented by the predicted bounding boxes.

Subsequently, AITABLER applies OCR to each detected region to convert visual content into machine-readable text, thereby preparing structured inputs for downstream semantic analysis. As a result, AITABLER converts raw PDF documents into structured representations, containing for each page the region types, textual content, and relative spatial information of all detected components.

3.2 Table-Text Semantic Relevance Inference

Spatial Context Retrieval. To assess the semantic association between a specific table and a text paragraph, a straightforward strategy is to directly

analyze their pairwise relationship [12]. However, prior studies have shown that natural language is inherently contextual and semantically dependent on surrounding discourse[3,30]. As a result, analyzing a paragraph in isolation may lead to misjudgment due to incomplete semantic cues.

To mitigate this issue, AITABLER augments the target paragraph with spatially proximate context. Leveraging block-level structural information from *Document Preprocessing* (Section 3.1), it selects surrounding text blocks based on their relative positions in the document’s reading order, thereby enriching semantic evidence for relevance assessment. Concretely, each block is represented by the geometric center of its bounding box as a spatial anchor. We then define a layout-aware *reading-order distance* to quantify proximity between blocks. Considering the distinct reading patterns of single- and double-column layouts, we adopt a unified formulation with layout-specific adaptations.

1. Reading-order distance for single-column layouts. In single-column documents, the reading order of coarse-grained blocks is unambiguously top-to-bottom within each page. For two blocks a and b located on the same page with vertical centers y_a and y_b , the reading-order distance is defined in Equation (Eq) (1):

$$d_{RO}(a, b) = |y_a - y_b|. \quad (1)$$

When the two blocks reside on adjacent pages p_a and p_b ($p_a < p_b$), we preserve reading continuity by concatenating their residual distances to the nearest page boundaries as shown in Eq. (2), where $y_{\text{bot}}^{p_a}$ and $y_{\text{top}}^{p_b}$ denote the bottom and top content boundaries of the respective pages. This formulation treats page breaks as continuous transitions in the reading stream.

$$d_{RO}(a, b) = (y_{\text{bot}}^{p_a} - y_a) + (y_b - y_{\text{top}}^{p_b}), \quad (2)$$

2. Column linearization for double-column layouts. Many academic documents adopt a double-column layout, in which vertical proximity alone does not faithfully reflect reading order: blocks within the same column are typically closer in reading order than adjacent-column blocks even when they share identical vertical coordinates.

To handle this, we linearize the two-dimensional layout into a one-dimensional reading-order coordinate $\hat{y}$ via a column-unfolding transformation. Consider an N -page double-column document whose $2N$ half-columns are indexed in reading order (left column, then right column, pages in ascending order), with H_j denoting the content height of column j . For a block b located in column $\kappa(b)$ at vertical position y_b , its linearized coordinate $\hat{y}(b)$ is computed as Eq. (3):

$$\hat{y}(b) = \sum_{j=1}^{\kappa(b)-1} H_j + (y_b - y_{\text{top}}^{\kappa(b)}). \quad (3)$$

Here, $y_{\text{top}}^{\kappa(b)}$ denotes the top content boundary of column $\kappa(b)$. By “stacking” all $2N$ columns into a single vertical axis, this transformation yields a unified reading-order distance across columns and pages. Accordingly, the reading-order distance

between two blocks a and b is computed as shown in Eq. (4). This definition naturally generalizes the single-column setting, which becomes a special case when each page contains only one column.

$$d_{\text{RO}}(a, b) = |\hat{y}(a) - \hat{y}(b)|. \quad (4)$$

3. *K-nearest-neighbour context retrieval.* With the reading-order distance d_{RO} defined, context expansion is formulated as a K -nearest-neighbour (K -NN) retrieval problem in the linearized reading-order space. With $\mathcal{B}$ denoting the set of candidate blocks within a configurable search scope, for a target table t , the context set $\mathcal{C}_K(t)$ is determined as Eq. (5). That is, $\mathcal{C}_K(t)$ consists of the K blocks with the smallest reading-order distance to t . Both the search scope $\mathcal{B}$ and the neighbourhood size K are configurable to accommodate different document characteristics. The retrieved context set $\mathcal{C}_K(t)$ is subsequently provided to the LLM for relevance grading.

$$\mathcal{C}_K(t) = \arg \min_{b \in \mathcal{B}} K d_{\text{RO}}(b, t). \quad (5)$$

Semantic Relevance Grading. With the grading scheme and spatial context retrieval established, we now turn to automating the semantic relevance grading for *table* and *text* blocks. We investigate two complementary approaches: an encoder-based classifier for lightweight local deployment and a context-enhanced LLM approach that forms the primary methodological contribution of this work.

1. *Encoder-based Classification.* We extend the Table-Text Association Model (TTAM) from prior work [12], which performs binary related/unrelated classification, to a five-class classifier. The architecture feeds the concatenated input [CLS] t [SEP] c [SEP], where t and c denote the content of the table and the candidate block, respectively, into a pretrained encoder (e.g., BERT [6]), and replaces the binary head with a five-way linear layer. Training uses cross-entropy loss with inverse-frequency class weights to mitigate the severe class imbalance.

2. *Context-enhanced LLM Grading.* We design an LLM-based grading method that prompts a large language model with the five-level criteria described in Section 2.1, combining the contextual information. Concretely, each query supplies the target table t and the candidate block c , and the model returns a predicted relevance level together with a textual justification and a confidence score. To improve grading accuracy, the K nearest blocks $\mathcal{C}_K(t)$ retrieved via the spatial context strategy described in Section 3.2 are supplied as auxiliary context, anchoring the LLM’s understanding of the table’s surroundings in the document. Figure 2 shows the prompt template, where the LLM is explicitly instructed to focus on the (t, c) pair and to use the context solely for disambiguation, not for direct comparison. Here, we apply spatial context augmentation only to the LLM-based approach, as encoder-based models are constrained by limited input lengths (typically a few hundred tokens, e.g., 256 for BERT), which restrict their ability to incorporate additional contextual information [6].

System: You are a semantic table–text relevance evaluator.

Task: Given a Target Table and one Candidate text block, classify their relevance into one of five levels.

Level 1 (Title Reference) – The candidate is semantically highly aligned with the table’s title or caption, establishing a direct referential link.

Level 2 (Content Alignment) – The candidate interprets, summarizes, or reasons about specific data in the table body.

Level 3 (Topical Match) – The candidate shares conceptual or thematic overlap with the table’s headers, but does not engage with specific data values.

Level 4 (Peripheral Context) – Indirect thematic connection; understanding the link requires background knowledge.

Level 5 (Unrelated) – No semantic relationship.

Decision procedure: Evaluate in descending order from Level 1 to Level 3. At each level, determine whether the candidate’s semantic relationship with the table satisfies the criterion; if evidence is insufficient, proceed to the next level. Assign Level 4 or 5 based on residual topical proximity.

Context: {context_blocks}

Target Table: {table_content}

Candidate: {candidate_text}

Output: Level ⟨1–5⟩ – Justification – Confidence

Fig. 2: Prompt template for context-enhanced five-level relevance grading. Placeholders in **braces** are filled at inference time.

3.3 Semantic-Enhanced Table Question Answering

Building upon the core reasoning capability of AITABLER, we further develop a table question answering module to support an important class of real-world question-answering tasks over document tables. Given a target PDF document and a table-related query, AITABLER first retrieves text paragraphs and their surrounding context that are semantically associated with the target table. Through fine-grained table–text relevance analysis, AITABLER returns context blocks with varying relevance levels and supplies them, together with the original question, to an LLM for joint reasoning. This design enables *semantic-enhanced* table question answering, where table reasoning is augmented by structured contextual evidence from the document.

3.4 Dataset: Semantic Grading & Table-QA Dataset Annotation

To the best of our knowledge, there has been no prior study systematically investigates fine-grained semantic relevance between tables and surrounding texts. To fill this gap and enable a rigorous evaluation, we construct a dedicated dataset. The construction procedure is described as follows.

Document Collection and Block Extraction. We collected documents from two open-access academic repositories: arXiv [5] (123 PDFs with diverse formatting templates) and PubMed Central [16] (103 PDFs from publishers

Table 1: Statistics of the table-context relevance grading dataset.

Source	#PDF	#Page	#Table	#Text	#Pair
arXiv	123	2,392	761	10,097	20,103
PubMed Central	103	1,544	412	4,923	9,035
Total	226	3,936	1,173	15,020	29,138

Table 2: Distribution of relevance levels in the grading dataset.

	L1 (Title)	L2 (Content)	L3 (Topical)	L4 (Peripheral)	L5 (Unrelated)
Count	800	617	1,918	11,415	14,388
(%)	2.7	2.1	6.6	39.2	49.4

including MDPI, Springer Nature, PLOS, and SAGE). Each document was processed through the *preprocessing pipeline* (Section 3.1) to produce *Table*, and *Text* blocks.

Grading Dataset. We treat all parsed blocks labeled as *Table* as anchors. For each table, all *text* blocks on the same page as well as adjacent pages are paired as candidate instances. All annotations were performed by a team of trained expert annotators following the five-level grading scheme defined in Section 2.1. To ensure labeling consistency, each table-text pair was independently annotated by two annotators; disagreements were resolved through discussion and adjudication by a senior annotator. Each candidate pair is recorded as a structured annotation: $\langle \text{Table-ID}, \text{Table-Page-ID}, (\text{Table-Coordinate}), \text{Text-ID}, \text{Text-Page-ID}, (\text{Text-Coordinate}), \text{Relevance Level} \rangle$, where coordinates are bounding boxes represented as $(y_{rb}, x_{rb}, y_{lu}, x_{lu})$ with *rb* and *lu* denoting the right-bottom and left-upper corners, respectively.

Table 1 summarizes overall dataset statistics, and Table 2 shows the distribution across five relevance levels. A clear monotonic trend emerges: the number of pairs increases as relevance decreases from Level 2 to Level 5, consistent with document structure that few paragraphs directly describe table content, more exhibit partial overlap, and most are semantically distant. The only exception is that Level 1 slightly exceeds Level 2, which is expected: title-based references are often repetitive, whereas Level 2 requires explicit engagement with table content. Overall, highly relevant pairs (Levels 1–2) account for only 4.9%, while peripheral and unrelated pairs (Levels 4–5) constitute 88.5%, highlighting the sparsity of strong table-text associations in real-world documents.

Table-QA Dataset. Using the same 226 PDFs and 1,173 tables, we construct a Table-QA benchmark with 6,000 questions. Following operation taxonomies from prior Table-QA benchmarks [18,33,2,23], we define five cognitive categories:

- **C1 (Direct Lookup)** extracts a single cell value (WTQ [18], TaPas [8]);
- **C2 (Arithmetic)** performs numerical operations such as sum, ratio, or percentage change (TAT-QA [33]);

Table 3: Composition of the Table QA benchmark by question type.

	C1	C2	C3	C4	C5	Total
Count	900	1,499	1,200	1,200	1,201	6,000
(%)	15	25	20	20	20	100

- **C3 (Comparison)** requires ranking or identifying extrema (SCITAT [27], TableBench [23]);
- **C4 (Multi-hop Reasoning)** chains multiple reasoning steps (HybridQA [3], MultiHiertt [30]);
- **C5 (Fact Verification)** evaluates statement truthfulness (TabFact [2]).

The distribution shown in Table 3 reflects practical table analysis demands: C1 accounts for 15% (baseline lookup capability), C2 for 25% (numerical reasoning dominates real-world analysis), and C3–C5 each for 20%, ensuring balanced coverage of comparison, multi-hop reasoning, and fact verification. These proportions are consistent with prior benchmarks, where arithmetic and multi-step reasoning often dominate [33,30].

We adopt a three-stage expert annotation pipeline [33,30]. First, question composers generate items based solely on the target table. Second, independent answerers respond using both the table and full PDF context; answers are restricted to short-form (1–5 tokens) and cross-validated for consensus. Third, finalized answers are returned to the original composer for approval. This procedure yields an inter-annotator agreement of 92.2% (exact match), comparable to FinQA [4] (92.7%) and MultiHiertt [30] ($\kappa=0.72\text{--}0.93$). All answers are short-form (number, text, or boolean), enabling fully automatic evaluation. Numeric answers are compared with tolerance $\varepsilon = 0.01$ after percentage/decimal normalization; text answers use case-insensitive exact match; and boolean answers are evaluated via binary accuracy.

4 Experiments

4.1 Implementation Details

Document Preprocessing. We adopt pdf2image [1] to convert PDF documents into page-level images. For layout parsing, we employ Faster R-CNN implemented within the Detectron2 [24] framework to perform structural detection on each page. The model is trained using the publicly available PubLayNet dataset [32]. This preprocessing stage produces structured blocks that serve as inputs for subsequent semantic modeling.

Model Implementation. Our experimental framework consists of two groups of models. First, we evaluate several encoder-based pretrained language models as lightweight baselines, including BERT [6], RoBERTa [14], and DeBERTa [7]. These models are fine-tuned for the relevance classification task. All model loading, training, and inference procedures are implemented using the Hugging Face Transformers library [10]. Second, we conduct experiments using a large

language model, GPT-5.2 [17]. Both zero-shot and structured prompting strategies are explored. LLM-based experiments are performed via API access.

Experimental Environment. All experiments, including training and evaluation, are conducted on an Ubuntu 22.04 server with an RTX 4090 GPU.

4.2 Evaluation Metrics

We evaluate grading performance with **Accuracy** and **F1-score**. Accuracy measures the fraction of correctly classified instances. F1-score is the harmonic mean of precision and recall. To aggregate across the five levels we report **Macro-averaged F1**, which averages the per-level F1-scores uniformly by assigning equal weight to every level regardless of its frequency. This metric prevents the overall score from being dominated by any single class.

4.3 Table-Context Relevance Grading Results

We evaluate two approaches to five-level table-context relevance grading on the dataset described in Table 1 (29,138 table-text pairs from 226 scientific PDFs). The first approach fine-tunes encoder-based classifiers with a 7:3 train/test split. The second approach directly prompts GPT-5.2 in a zero-shot setting. To isolate the contribution of spatial context, we run this approach both *without* context (table-text pair only) and *with* context, where up to 22 spatially closest components from the current and adjacent pages are supplied as auxiliary input. For fair comparison, we report LLM results on the same held-out test set; we additionally evaluate GPT-5.2 on the full dataset (marked with * in Table 4) to characterise its behaviour at scale. Table 4 summarises all results in terms of per-level accuracy, per-level F1, Macro F1, and overall accuracy.

Grading by Encoder-based Models. As shown in Table 4, we finetuned BERT, RoBERTa, and DeBERTa to evaluate the effectiveness of encoder-based approaches for fine-grained table-text relevance classification, which constitutes the primary strategy adopted by DoTABLER [12].

However, unlike binary related/unrelated classification, encoder-based models demonstrate limited capability in distinguishing fine-grained relevance levels. Except for the strongly related L1 category, the classification accuracy of all three models remains below 60% across the other levels. Further analysis suggests that this performance limitation stems from the intrinsic difficulty of fine-grained relevance modeling. Encoder-based architectures are less effective at capturing subtle semantic distinctions and long-range contextual dependencies, making them prone to confusion between adjacent relevance levels.

LLM Grading on the Test Set. Our results demonstrate that current LLMs possess a strong capability for fine-grained semantic relevance assessment between tables and their surrounding text, highlighting the potential of zero-shot LLMs for multi-level relevance assessment in document understanding tasks.

With context enhancement, GPT-5.2 achieves a Macro F1 of 70.82% and an overall accuracy of 73.13% on the held-out test set. For Levels 1–3, which

Table 4: Relevance grading performance. “+ ctx” denotes context-enhanced grading with top-22 nearby components.

Method	Metric (%)	Per-Level					Macro	Overall
		L1	L2	L3	L4	L5	F1	Acc
BERT	Acc	89.15	52.16	37.00	37.23	58.17	41.87	47.06
	F1	33.48	43.28	28.35	46.78	57.51		
RoBERTa	Acc	83.72	56.17	24.57	35.82	51.10	36.15	42.72
	F1	27.20	34.15	21.07	45.64	52.68		
DeBERTa	Acc	88.37	43.21	35.71	43.57	47.24	40.01	45.66
	F1	28.93	41.67	26.12	51.58	51.73		
GPT-5.2	Acc	98.12	98.79	94.03	70.28	75.21	62.64	67.12
	F1	53.48	66.12	53.21	65.68	74.72		
GPT-5.2 + ctx	Acc	98.58	99.41	95.43	75.52	78.69	70.82	73.13
	F1	69.12	71.38	62.63	73.41	77.58		
GPT-5.2*	Acc	97.47	98.12	92.91	69.22	74.25	60.29	65.97
	F1	51.49	63.50	51.43	63.23	71.80		
GPT-5.2 + ctx*	Acc	<u>97.83</u>	<u>98.50</u>	<u>94.38</u>	<u>74.51</u>	<u>77.82</u>	<u>68.14</u>	<u>71.52</u>
	F1	<u>66.29</u>	<u>68.87</u>	<u>60.11</u>	<u>70.97</u>	<u>74.46</u>		

Bold: best on test set; underline: best on full set. *Evaluated on the full dataset.

are defined by explicit structural cues such as caption reference, cell content alignment, and header overlap, the model reaches over 95% per-level accuracy. This indicates that GPT-5.2 can reliably identify fine-grained associations when the grading criteria are clearly grounded in observable table components. In contrast, Level 4 (Peripheral Context) remains the most challenging. Unlike the first three levels, L4 involves indirect thematic relevance without clear structural anchors. As a result, the boundary between peripheral relevance and unrelated content becomes less distinct, making automated grading more difficult and requiring broader semantic judgment.

Comparing the context-enhanced variant (+ ctx) with the table-text-only baseline further demonstrates the benefit of spatial context, with Macro F1 improving by **+8.18 percentage points (pp)**. The impact of context varies across levels in a pattern consistent with headroom: for L1–L3, per-level accuracy already exceeds 94.03% under the no-context setting, leaving limited room for further gains; context enhancement therefore yields modest accuracy improvements on these levels. For L4 and L5, however, where baseline accuracy is considerably lower, context provides substantially larger improvements. Besides, L4 accuracy rises from 70.28% to 75.52% (+5.24 pp) and L5 from 75.21% to 78.69% (+3.48 pp), confirming that spatial context is particularly valuable for disambiguating the more semantically ambiguous relevance boundaries.

Full-Dataset LLM Evaluation. The * rows in Table 4 report GPT-5.2 results on the entire 29,138-pair dataset. Context enhancement remains effective at this scale, yielding a Macro F1 improvement of **+7.85 percentage points**. These improvements closely mirror the trends observed on the test set, indicating that

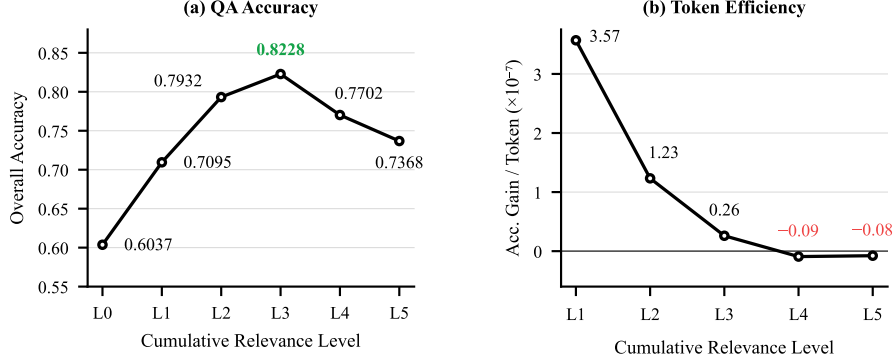


Fig. 3: (a) QA accuracy under cumulative relevance levels (L0 = table only). (b) Token efficiency: accuracy gain per additional token ($\times 10^{-7}$) at each level.

the impact of spatial context is consistent across evaluation settings. Although absolute scores are slightly lower due to the greater diversity of table-text configurations in the full dataset, the consistent performance gains demonstrate that the benefits of spatial context generalise beyond the held-out partition.

4.4 Impact of Graded Context on Table-QA

To evaluate whether our graded relevance annotations translate into tangible downstream benefits, we conduct a table-centric QA experiment on the QA benchmark proposed in this paper (Table. 3), which is built upon our graded relevance dataset (Table 1 and Table 2). We prompt GPT-5.2 to answer questions under progressively richer context configurations, starting from the table alone (L0) and cumulatively adding paragraphs of increasing relevance levels (L1, L1-2, up to L1-5). This setup allows us to (1) verify that higher-relevance paragraphs contribute larger accuracy improvements, and (2) quantify the marginal utility of each relevance level per additional token, thereby evaluating the cost-effectiveness gradient of our graded context selection. We measure overall QA accuracy as the fraction of correctly answered questions over the total. To assess cost-effectiveness, we define the token efficiency (TE) of relevance level L_i :

$$\text{TE}(L_i) = \frac{\text{Acc}(C_i) - \text{Acc}(C_{i-1})}{\text{Tokens}(L_i)}, \quad (6)$$

where C_i denotes the context configuration that cumulatively includes all paragraphs up to level L_i , and $\text{Tokens}(L_i)$ is the number of additional prompt tokens introduced by the L_i paragraphs ($i \in \{1, \dots, 5\}$). Intuitively, TE measures the accuracy gain per additional token of prompt, providing a practical indicator for controlling LLM inference cost under a given context budget.

Fig. 3(a) and Table 5 show that cumulatively incorporating paragraphs from L1 through L3 produces substantial and consistent accuracy improvements. Starting from a table-only baseline of 60.37%, adding L1 paragraphs raises

Table 5: QA accuracy under cumulative relevance levels. L0 uses the table alone; each subsequent level cumulatively adds paragraphs graded at that level.

	L0	L1	L2	L3	L4	L5
Accuracy (%)	60.37	70.95	79.32	82.28	77.02	73.68
Δ Acc (pp)	—	+10.58	+8.37	+2.96	−5.26	−3.34

Table 6: Token efficiency of each relevance level. Δ Tokens is the number of additional prompt tokens introduced by adding level- L_i paragraphs. Negative TE values indicate accuracy degradation.

	L1	L2	L3	L4	L5
Total Tokens	4.88M	5.56M	6.70M	12.42M	16.75M
Δ Tokens	296K	679K	1,136K	5,726K	4,332K
Δ Acc (pp)	+10.58	+8.37	+2.96	−5.26	−3.34
TE ($\times 10^{-7}$)	3.57	1.23	0.26	−0.09	−0.08

accuracy by 10.58 pp to 70.95%. Further including L2 and L3 increases accuracy to 79.32% and 82.28%, respectively, with a total gain of 21.9 pp over the table-only baseline. This confirms that for Table-QA, paragraphs with high relevance provide essential reference information that cannot be derived from the table itself, such as definitions, background, and data interpretations.

The token efficiency analysis in Fig. 3(b) and Table 6 further validates the effectiveness of our grading strategy. L1 paragraphs, which directly describe or reference the table, deliver the highest information density with a token efficiency of 3.57×10^{-7} , nearly $14\times$ that of L3 (0.26×10^{-7}), indicating the grading framework successfully identifies the most informative context first. L2 and L3 also yield positive returns (1.23×10^{-7} and 0.26×10^{-7}), though at progressively higher token costs. In practical terms, this means that AITABLER can prioritise the top three relevance tiers and still obtain the highest accuracy while consuming significantly fewer tokens than a naïve “include everything” strategy.

Beyond L3, performance degrades: incorporating L4–L5 reduces accuracy from 82.28% to 73.68% and yields negative token efficiency. This exposes a gap between relevance and utility: although L4 paragraphs are topically related, they rarely provide question-supporting evidence and instead act as distractors, while L5 further dilutes the signal through noise. The results confirm that more context does not necessarily improve LLM reasoning; low-relevance text can actively harm QA performance. These findings highlight the dual role of multi-level relevance grading: it forms a cost–effectiveness gradient (L1 $\rightarrow$ L3) that steadily improves accuracy, and acts as a filter against information overload. The optimal L1–L3 setting achieves 82.28% accuracy, outperforming the full-context baseline (73.68%) by 8.6 pp, while reducing prompt size by **60%** (6.70M vs. 16.75M tokens). From a deployment perspective, graded relevance enables controllable accuracy–cost trade-offs by prioritizing task-relevant context, reducing inference cost without sacrificing yet tending to improve performance.

5 Discussion

Model Effectiveness and Comparative Analysis. Experimental results show that large language models consistently outperform fine-tuned small encoders on multi-level semantic relevance grading, with additional gains from incorporating surrounding context. Although the grading taxonomy is explicitly defined, the task requires alignment with fine-grained semantic criteria rather than lexical matching. Small models rely solely on supervised data to approximate decision boundaries, limiting their ability to generalize abstract grading principles. In contrast, large language models leverage extensive pretraining and prompt-based conditioning to apply explicit grading rules during inference, enabling more reliable reasoning over nuanced relational signals. The consistent benefit of contextual augmentation further highlights the importance of coherent discourse integration, a capability better supported by large language models.

Limitations and Future Work. Despite its strong performance, AITABLER has limitations. It depends on external layout analysis and OCR pipelines, and preprocessing errors in highly complex or low-quality documents may affect downstream grading. However, this reflects limitations of document digitization pipelines rather than the semantic modeling strategy itself. Future work will focus on improving robustness to noisy or irregular documents, evaluating the framework on large-scale industrial datasets, and developing domain-adaptive semantic grading strategies.

Ethical Consideration. All documents were sourced from publicly accessible repositories (e.g., arXiv and PubMed Central Open Access) under open-access licenses (e.g., CC-BY). Only public content and metadata were used. No personal or sensitive data were involved, and all annotations were created solely for academic research without human subjects.

6 Conclusion

In this paper, we present AITABLER, a multi-level table–context relevance grading framework for structure-aware document understanding. By incorporating layout-aware distance and K -nearest-neighbour context selection, we inject spatial structure into LLM-based inference. Experiments show that spatially grounded context consistently improves grading performance across both held-out and full-dataset settings. Beyond grading, we demonstrate that fine-grained relevance enables more effective Table-QA: prioritizing higher-relevance context yields monotonic accuracy gains while improving token efficiency. Overall, our results establish multi-level relevance grading as a principled and practical mechanism for controllable context selection in table-centric document intelligence.

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